

Chan Park

AI Researcher | Vision-Language-Action Models for Driving and Robotics

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Summary Research spans driving VLA, closed-loop evaluation, robot-policy safety, and robust multimodal perception.

Education

Sungkyunkwan University M.S. in Artificial Intelligence; DASH Lab; Advisor: Prof. Simon S. Woo.	Mar. 2024 – Aug. 2026
University of Toronto Visiting graduate student via CARTE, in partnership with LG Electronics Toronto AI Lab. Mechanical and Industrial Engineering; Advisor: Prof. Steve Mann.	Jan. 2026 – Jul. 2026
Inha University B.S. in Industrial Engineering.	Mar. 2017 – Aug. 2023

Experience

LG Electronics, Toronto AI Lab <i>AI Research Intern, Robot Learning / VLA Safety</i> [Project Page]	Jan. 2026 – Aug. 2026
- Improved collision avoidance from 64.7% to 69.1% (average over 4 SafeLIBERO suites, vs. AEGIS) with Route-Guided CBF-QP Repair: soft route guidance and hard safety constraints; ablated scoring and rejoin guidance. - Built MuJoCo / RoboCasa simulations, expanded VR demonstrations with MimicGen, and LoRA fine-tuned OpenVLA, $\pi 0.5$, DreamVLA, and DD-VLA with multi-GPU Slurm training.	
DASH Lab, Sungkyunkwan University <i>Graduate Researcher, Multimodal Learning / Deepfake Detection</i> [Project Page]	Mar. 2024 – Dec. 2025
- Maintained $\geq 92.9\%$ video accuracy across all 8 DF-TCB temporal corruptions on FF++ (97.9% clean) with ICR-Net's frame-reliability prediction and selective correction; built DF-TCB on FaceForensics++ and DFDC. - Led working-level execution of IITP deepfake R&D; deployed Dockerized detection REST APIs to the Korean National Police Agency and Supreme Prosecutors' Office.	
Voice AI Lab, Inha University <i>Undergraduate Research Assistant</i>	Aug. 2022 – Aug. 2023
Built synchronized EMG / EEG / audio / video acquisition for silent-speech research (EMG 1,000 Hz; audio 16 kHz).	

Projects

Driving VLA: Unobservable Claims in Language-Conditioned Driving <i>Independent Research SimLingo / CARLA</i>	Sep. 2026 – Present
- Identified camera-unobservable causes in 5.48% of cause-bearing records (up to 38.1% for signalized-junction right turns) by auditing all 2,085,459 auto-generated commentary records. - Measured +0.85 m/s predicted-speed change by swapping cause clauses (open-loop; 95% route-bootstrap CI [0.69, 1.02]; 300 frames / 119 routes); identical-text reinjection changed speed by 0.000. - Measured 40.7% caution loss for deletion vs. 2.5% for hedged rewriting (open-loop; 204 hidden-hazard braking frames); the gap persisted under placebo edits. - Built CARLA / Bench2Drive closed-loop evaluation infrastructure with CHAIR scoring; results pending.	
LLM Agent Research Pipeline with Cross-Model Review <i>Personal project</i>	Sep. 2026
- Built a Python cross-model-family reviewer harness on the open-source ARIS skill pack for Claude Code: env-file API keys, multi-model fallback with back-off, provenance headers, and Codex CLI as second reviewer. - Flash-tier Gemini scored 8.2 / 7.63 (revise; provenance showed the 8.2 came from a fallback model) vs. Codex's 5.35 / 5.65 (no pass); kept objections open and caught a reviewer error (≥ 2048 vs. actual 896).	
Submersivity: XR-Guided Underwater Debris Detection <i>Mann Lab, University of Toronto</i> [Project Page]	Jan. 2026 – Jul. 2026
- Raised TACO validation mAP50 from 0.27 to 0.49 via single-class training; reached 92.1% precision after fine-tuning on 12,177 terrestrial and underwater images. - Integrated submarine video, YOLOv8s, COLMAP + ArUco, and Vuzix Smart Swim guidance; built the Android XR receiver with packet reassembly, 500 ms stale-data detection, and reconnection.	

- Built a lip-synthesis pipeline for 436 of 590 silent videos: visual speech recognition (Auto-AVSR), LLM word flips, Kokoro TTS, and MuseTalk lip sync, swapping only the altered-word frames located by Montreal Forced Aligner.
- Routed the 154 videos with unreliable transcripts to an identity-swap branch (ArcFace frame selection + SimSwap, 1–3 spliced segments of 15–45 frames); earned the Excellence Award (Jul. 2025).

Undergraduate Projects

- Built the **3D Figure Generation Platform** with PIFuHD as PM/back-end developer. [Project Page] Apr. 2023
- Identified 59 suitable sites among 168 rest areas in **Hydrogen Station Site Evaluation**. Fall 2022
- Fine-tuned MobileNet and built UI/deployment for the **Korean Food Classification App**. Jul. 2022 – Sep. 2022
- Finished the **Autonomous RC-Car Competition** with the lightest completing car. Fall 2017

Publications

1. **Chan Park**. “Salience, Not Visibility: Privileged Supervision Puts Unobservable Claims into a Driving VLA’s Control Loop.” Manuscript in preparation (target: NAACL).
2. **Chan Park**, Sunwoo Hong, Kirak Kim, Yoonjeong Park, Alexander W. Olson, and Ali Pesaranghader. “Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models.” Workshop on Trustworthy Embodied Foundation Models, Robotics: Science and Systems (RSS) 2026. [Project Page] [Code]
3. Chris McGale, Michel A. Herrera Viyella, **Chan Park**, Daniel Bros, Alexander Vicol, and Steve Mann. “Submersivity: XR-Guided Garbage Detection with an RC Submarine for Peter Street Basin Cleanup.” IEEE International Conference on Mechatronics and Automation (ICMA) 2026. [Project Page]
4. Alexander Vicol, Steve Mann, Chris McGale, Gianella Bejar-Alvarez, Michel Herrera Viyella, Darya Zanzanpour, **Chan Park**, Xiaoming Chen, Xueqi Yang, and Bingxuan Yang. “State-of-Flight: Wearable EEG and Motion Context During Flying or Floating.” IEEE International Symposium on Computer-Based Medical Systems (CBMS) 2026.
5. **Chan Park**, Hyeongjun Choi, Muhammad Shahid Muneer, Binh Minh Le, and Simon S. Woo. “ICR-Net: Robust Deepfake Detection under Temporal Corruption.” Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2026. [Paper] [Code]
6. **Chan Park**, Muhammad Shahid Muneer, and Simon S. Woo. “Beyond Masking: Landmark-based Representation Learning and Knowledge-Distillation for Audio-Visual Deepfake Detection.” ACM International Conference on Information and Knowledge Management (CIKM) 2025, Short Paper. [Paper] [Code]
7. **Chan Park***, Bohyun Moon*, Minsun Jeon, Jee-weon Jung, and Simon S. Woo. “X3A: Efficient Multimodal Deepfake Detection with Score-Level Fusion.” ACM/SIGAPP Symposium on Applied Computing (SAC) 2025. [Paper]
8. “Weighted Contrastive Learning for Robust Audio-Visual Deepfake Detection.” Under review. [Code]

* equal contribution

Patents & Software

Method and Apparatus for Deepfake Detection under Temporal Corruption (Korean patent application filed) 2026
Efficient Multimodal Deepfake Detection with Score-Level Fusion (Korean software copyright registration) 2025

Honors & Awards

Korean Government Scholarship | IITP AI Excellence Global Innovative Leader Education Program (MSIT) 2026
Top 10% | 1M-Deepfakes Detection Challenge, ACM Multimedia 2025 [Challenge] 2025
Excellence Award | KIISE Deepfake Generation & Detection Competition (Team MetaForensic) Jul. 2025
SKKU AI Research Fellowship; BK21 Master’s Research Fellowship | Sungkyunkwan University 2024–2026
Undergraduate Research Fellowship (×2), Inha University; City-funded Future Talent Scholarship, Siheung City.

Skills

Programming: Python, C++, Java (Android), SQL; Docker, REST APIs (Flask), Linux, Git, cloud GPU VMs.
ML / CV: PyTorch, LoRA fine-tuning, multi-GPU training (Slurm), OpenCV, YOLOv8, Hugging Face.
Simulation / Robotics: CARLA 0.9.15, Bench2Drive, MuJoCo, RoboCasa, MimicGen, VR teleoperation.
LLM / Agents: Claude Code, Codex CLI, Gemini API, MCP.
Languages: Korean (native); English (OPIc Advanced Low, Sep. 2026).